

Extended 2D Fokker–Planck Solver: $\mathcal{M} = \{(1, 0), (1, 1)\}$ Call Nr. 7; Notes about the code

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1 Model Equations

1.1 Fokker–Planck PDE

Following the given formulation, the cluster density $U(x, y, t)$ satisfies

$$\frac{\partial U}{\partial t} = -\frac{\partial(F_x U)}{\partial x} - \frac{\partial(F_y U)}{\partial y} + \frac{\partial^2(D_{xx} U)}{\partial x^2} + \frac{\partial^2(D_{yy} U)}{\partial y^2} + \frac{\partial^2(D_{xy} U)}{\partial x \partial y}, \quad (1)$$

where, for a mobile set $\mathcal{M}$ with species $\mu = (m, q)$ and concentrations C_μ ,

$$F_x = \sum_{\mu \in \mathcal{M}} m (\beta_\mu C_\mu - \alpha_\mu), \quad F_y = \sum_{\mu \in \mathcal{M}} q (\beta_\mu C_\mu - \alpha_\mu), \quad (2)$$

$$D_{xx} = \frac{1}{2} \sum_{\mu \in \mathcal{M}} m^2 (\beta_\mu C_\mu + \alpha_\mu), \quad D_{yy} = \frac{1}{2} \sum_{\mu \in \mathcal{M}} q^2 (\beta_\mu C_\mu + \alpha_\mu), \quad D_{xy} = \sum_{\mu \in \mathcal{M}} m q (\beta_\mu C_\mu + \alpha_\mu). \quad (3)$$

1.2 Mobile set and kinetic rates

We use the mobile set $\mathcal{M} = \{(1, 0), (1, 1)\}$, where $(1, 0)$ moves only in x and $(1, 1)$ moves in both x and y . The parameter $\varepsilon \in [10^{-2}, 10^2]$ controls the relative mobility of species $(1, 1)$. In addition we use a constant source term $G(1, 0)$ which can be chosen between 10^{23} and 10^{17} . For $x \geq 2$:

$$\begin{aligned} \beta_{(1,0)}(x) &= x^{1/2}, & \beta_{(1,1)}(x) &= \varepsilon x^{1/2}, \\ E_{(1,0)}(x) &= 24.47 - 34 \left(x^{2/3} - (x-1)^{2/3} \right), & E_{(1,1)}(x) &= 27.47 - 34 \left(x^{2/3} - (x-1)^{2/3} \right), \\ \alpha_{(1,0)}(x) &= \beta_{(1,0)}(x) e^{-E_{(1,0)}(x)}, & \alpha_{(1,1)}(x) &= \beta_{(1,1)}(x) e^{-E_{(1,1)}(x)}. \end{aligned}$$

Since both β and α depend only on x (not y), the coefficients $F_x, F_y, D_{xx}, D_{yy}, D_{xy}$ are functions of x only. We write them as vectors indexed by the x -grid: $F_x[i], F_y[i]$, etc.

For diffusion we define the per-species kinetic amplitudes

$$d_1[i] = \max(\beta_{(1,0)}(x_i) C_{10} + \alpha_{(1,0)}(x_i), 0), \quad d_2[i] = \max(\beta_{(1,1)}(x_i) C_{11} + \alpha_{(1,1)}(x_i), 0),$$

2 Mobile Concentration Dynamics

The concentrations of C_{10} and C_{11} will change over time. So far we ignored this, but the constant absorption and emission of them is implicitly done in the model. Now we finally incorporate them. For this we do the following calculations:

2.1 Absorption integrals ϕ_{10} and ϕ_{11}

The B/A decomposition. Because β_μ and α_μ depend only on x (not on y or C_μ), the integral ϕ_μ is linear in C_μ . We split it into an *absorption coefficient* B_μ and an *emission integral* A_μ , both of which depend on U but *not* on the concentration C_μ :

$$B_\mu = \iint \beta_\mu(x) U(x, y) dx dy, \quad (4)$$

$$A_\mu = \iint \alpha_\mu(x) U(x, y) dx dy. \quad (5)$$

Then

$$\phi_\mu = B_\mu C_\mu - A_\mu. \quad (6)$$

B_μ is the rate at which the cluster lattice absorbs mobile species per unit concentration; A_μ is the rate at which it emits them unconditionally. Because both $\beta_\mu \geq 0$ and $\alpha_\mu \geq 0$, we have $B_\mu, A_\mu \geq 0$ whenever $U \geq 0$.

Discrete form. On the grid, column masses¹ $m_i = \Delta y \sum_j U_{ij}$ give

$$B_{10} = \Delta x \sum_{i=1}^{N_x} \beta_{(1,0)}(x_i) m_i, \quad A_{10} = \Delta x \sum_{i=1}^{N_x} \alpha_{(1,0)}(x_i) m_i, \quad (7)$$

$$B_{11} = \Delta x \sum_{i=1}^{N_x} \beta_{(1,1)}(x_i) m_i, \quad A_{11} = \Delta x \sum_{i=1}^{N_x} \alpha_{(1,1)}(x_i) m_i, \quad (8)$$

so that $\phi_{10} = B_{10}C_{10} - A_{10}$ and $\phi_{11} = B_{11}C_{11} - A_{11}$.

2.2 Update equations for C_{10} and C_{11}

The mobile concentrations obey

$$\frac{dC_{10}}{dt} = G_{10} - \phi_{10} = G_{10} + A_{10} - B_{10} C_{10}, \quad (9)$$

$$\frac{dC_{11}}{dt} = -\phi_{11} = A_{11} - B_{11} C_{11}, \quad (10)$$

where $G_{10} \geq 0$ is an external source for species $(1, 0)$ and there is no source for $(1, 1)$. When $B_\mu C_\mu > A_\mu$ the lattice is a net absorber (reducing C_μ). This explains why in some experiments the concentration values fade out over time (which looks like mass-conservation is violated). When $B_\mu C_\mu < A_\mu$ the lattice emits into the mobile phase (increasing C_μ).

2.3 Material budgets X_{mat} and Y_{mat}

The Fokker-Planck equation for U is in flux form with zero-flux boundary conditions, so the mass $M_U = \iint U dx dy$ is conserved exactly by the PDE². The mobile concentrations do change, and the combined system satisfies two global budget quantities:

$$Y_{\text{mat}}(t) = C_{11}(t) + \iint y U(x, y, t) dx dy, \quad (11)$$

$$X_{\text{mat}}(t) = C_{10}(t) + C_{11}(t) + \iint x U(x, y, t) dx dy. \quad (12)$$

¹As all values along y are the same for β and α we can already sum the mass up.

²This is also enforced numerically by using finite volume schemes.

Derivation of how X_{mat} and Y_{mat} change. Write (1) in flux form $\partial_t U = -\partial_x J_x - \partial_y J_y$ with zero-flux boundary conditions ($J_x = J_y = 0$ on $\partial\Omega$). Multiplying by y and integrating by parts in y :

$$\frac{d}{dt} \iint y U \, dx \, dy = \iint y (-\partial_x J_x - \partial_y J_y) \, dx \, dy = \underbrace{\iint y (-\partial_x J_x) \, dx \, dy}_{=0} + \iint y (-\partial_y J_y) \, dx \, dy.$$

To see why the first integral vanishes explicitly, we factor out y as it is independent of x :

$$\int y \left(\int -\partial_x J_x \, dx \right) dy = \int y [-J_x]_{\partial_x \Omega} dy = 0,$$

which evaluates to zero because $J_x = 0$ at the x -boundaries. The second integral uses integration by parts in y with $[y J_y]_{\partial_y \Omega} = 0$, yielding $\iint J_y \, dx \, dy$. The y -flux is $J_y = F_y U - (\text{diffusion})$; under zero-flux BCs the diffusion terms integrate to zero, leaving

$$\frac{d}{dt} \iint y U \, dx \, dy = \iint F_y U \, dx \, dy.$$

For $\mathcal{M} = \{(1,0), (1,1)\}$, the coefficient $F_y = \beta_{(1,1)} C_{11} - \alpha_{(1,1)}$. Expanding this explicitly gives exactly ϕ_{11} (using Eq. 6):

$$\iint F_y U \, dx \, dy = \iint (\beta_{(1,1)} C_{11} - \alpha_{(1,1)}) U \, dx \, dy = \phi_{11}.$$

Therefore, the full time derivative of Y_{mat} is:

$$\frac{dY_{\text{mat}}}{dt} = \frac{d}{dt} \left(C_{11}(t) + \iint y U \, dx \, dy \right) = \underbrace{\frac{dC_{11}}{dt}}_{-\phi_{11}} + \underbrace{\iint F_y U \, dx \, dy}_{\phi_{11}} = 0.$$

Applying the exact same expansion to X_{mat} , we multiply the flux form by x and integrate over the domain:

$$\frac{d}{dt} \iint x U \, dx \, dy = \iint x (-\partial_x J_x - \partial_y J_y) \, dx \, dy = \iint x (-\partial_x J_x) \, dx \, dy + \underbrace{\iint x (-\partial_y J_y) \, dx \, dy}_{=0}.$$

The second integral evaluates to zero because x is independent of y , allowing us to factor it out:

$$\int x \left(\int -\partial_y J_y \, dy \right) dx = \int x [-J_y]_{\partial_y \Omega} dx = 0,$$

which vanishes because $J_y = 0$ at the y -boundaries. For the first integral, we use integration by parts in x with $[x J_x]_{\partial_x \Omega} = 0$, yielding $\iint J_x \, dx \, dy$. The x -flux is $J_x = F_x U - (\text{diffusion})$; under zero-flux BCs the diffusion terms integrate to zero, leaving

$$\frac{d}{dt} \iint x U \, dx \, dy = \iint F_x U \, dx \, dy.$$

For $\mathcal{M} = \{(1,0), (1,1)\}$, the coefficient F_x receives contributions from both species: $F_x = \beta_{(1,0)} C_{10} - \alpha_{(1,0)} + \beta_{(1,1)} C_{11} - \alpha_{(1,1)}$. Expanding this integral explicitly yields $\phi_{10} + \phi_{11}$:

$$\iint F_x U \, dx \, dy = \iint (\beta_{(1,0)} C_{10} - \alpha_{(1,0)}) U \, dx \, dy + \iint (\beta_{(1,1)} C_{11} - \alpha_{(1,1)}) U \, dx \, dy = \phi_{10} + \phi_{11}.$$

Finally, we take the full time derivative of X_{mat} and substitute the ODEs for C_{10} and C_{11} :

$$\begin{aligned}
\frac{dX_{\text{mat}}}{dt} &= \frac{d}{dt} \left(C_{10} + C_{11} + \iint x U \, dx \, dy \right) \\
&= \frac{dC_{10}}{dt} + \frac{dC_{11}}{dt} + \frac{d}{dt} \iint x U \, dx \, dy \\
&= \underbrace{(G_{10} - \phi_{10})}_{\dot{C}_{10}} + \underbrace{(-\phi_{11})}_{\dot{C}_{11}} + \underbrace{(\phi_{10} + \phi_{11})}_{\iint F_x U \, dx \, dy} \\
&= G_{10}.
\end{aligned}$$

These quantities are monitored in the simulation output to verify that the coupling between U and (C_{10}, C_{11}) is implemented correctly.

3 Specialisation to $\mathcal{M} = \{(1, 0), (1, 1)\}$

3.1 Diffusion Tensor Derivation

The tensor coefficients are

$$D_{xx} = \frac{1}{2} \sum_{\mu \in \mathcal{M}} m^2 d_\mu, \quad D_{yy} = \frac{1}{2} \sum_{\mu \in \mathcal{M}} q^2 d_\mu, \quad D_{xy} = \sum_{\mu \in \mathcal{M}} m q d_\mu.$$

For $\mathcal{M} = \{(1, 0), (1, 1)\}$ with $d_{(1,0)} = d_1$, $d_{(1,1)} = d_2$:

$$\begin{aligned}
D_{xx}[i] &= \frac{1}{2} ((1)^2 d_1[i] + (1)^2 d_2[i]) = \frac{d_1[i] + d_2[i]}{2}, \\
D_{yy}[i] &= \frac{1}{2} ((0)^2 d_1[i] + (1)^2 d_2[i]) = \frac{d_2[i]}{2}, \\
D_{xy}[i] &= (1 \cdot 0) d_1[i] + (1 \cdot 1) d_2[i] = d_2[i].
\end{aligned}$$

Therefore

$$\boxed{D_{xy}[i] = 2D_{yy}[i]}.$$

3.2 Advection terms

From (2), specialising to $\mathcal{M} = \{(1, 0), (1, 1)\}$:

$$\begin{aligned}
F_x[i] &= (\beta_{(1,0)}(x_i) C_{10} - \alpha_{(1,0)}(x_i)) + (\beta_{(1,1)}(x_i) C_{11} - \alpha_{(1,1)}(x_i)), \\
F_y[i] &= \beta_{(1,1)}(x_i) C_{11} - \alpha_{(1,1)}(x_i).
\end{aligned}$$

Note that F_x and F_y depend only on x (not on y), so they are stored as 1D vectors indexed by the x -grid. The conservative advection operator is

$$\text{Adv}(U) = -\partial_x(F_x U) - \partial_y(F_y U),$$

discretised with MP5 reconstruction and SSP-RK3 time integration.

4 Mesh-Adapted Operator Splitting

The orange part was copied from last week's notes and mostly unchanged.

4.1 Diagonal finite-difference stencil for general $\Delta x, \Delta y$

In this subsection, U denotes a generic scalar field sampled on the grid. In the diffusion split below we apply the same stencil either to $U = C$ (constant coefficients) or, locally, to coefficient-weighted quantities such as $U = \tilde{D}_2 C$.

A symmetric Taylor expansion about (x_i, y_j) gives:

$$U_{i+1,j+1} = U_{i,j} + \Delta x \partial_x U + \Delta y \partial_y U + \frac{\Delta x^2}{2} \partial_{xx} U + \Delta x \Delta y \partial_{xy} U + \frac{\Delta y^2}{2} \partial_{yy} U + \mathcal{O}(h^3), \quad (13)$$

$$U_{i-1,j-1} = U_{i,j} - \Delta x \partial_x U - \Delta y \partial_y U + \frac{\Delta x^2}{2} \partial_{xx} U + \Delta x \Delta y \partial_{xy} U + \frac{\Delta y^2}{2} \partial_{yy} U + \mathcal{O}(h^3), \quad (14)$$

where $h = \max(\Delta x, \Delta y)$. Adding the two expansions³ and rearranging gives:

$$U_{i+1,j+1} + U_{i-1,j-1} - 2U_{i,j} = \Delta x^2 \partial_{xx} U + 2\Delta x \Delta y \partial_{xy} U + \Delta y^2 \partial_{yy} U + \mathcal{O}(h^4). \quad (15)$$

Dividing through by $\Delta x \Delta y$:

$$\boxed{\frac{U_{i+1,j+1} - 2U_{i,j} + U_{i-1,j-1}}{\Delta x \Delta y} = \frac{\Delta x}{\Delta y} \partial_{xx} U + 2 \partial_{xy} U + \frac{\Delta y}{\Delta x} \partial_{yy} U + \mathcal{O}(h^2).} \quad (16)$$

$$\mathcal{T}_h \phi := r \partial_{xx} \phi + 2 \partial_{xy} \phi + \frac{1}{r} \partial_{yy} \phi, \quad r := \frac{\Delta x}{\Delta y}. \quad (17)$$

4.2 Mesh-adapted operator splitting

We seek $\tilde{A}_x, \tilde{A}_y, \tilde{D}_2$ such that

$$\text{Diff}(C) = \partial_{xx}(\tilde{A}_x C) + \partial_{yy}(\tilde{A}_y C) + \tilde{D}_2 \mathcal{T}_h(C), \quad (18)$$

After $\mathcal{T}_h$ via (17) and reorder the derivatives, we want to find original formulation again:

$$\partial_{xx}(\tilde{A}_x + r \tilde{D}_2)C + \partial_{yy}\left(\tilde{A}_y + \frac{1}{r} \tilde{D}_2\right)C + 2\tilde{D}_2 \partial_{xy} C = \partial_{xx}(D_{xx}C) + \partial_{yy}(D_{yy}C) + \partial_{xy}(D_{xy}C). \quad (19)$$

Now match coefficients of $\partial_{xx}, \partial_{yy}, \partial_{xy}$:

$$\partial_{xy} : \quad 2\tilde{D}_2 = D_{xy} \implies \tilde{D}_2 = \frac{D_{xy}}{2}, \quad (20)$$

$$\partial_{xx} : \quad \tilde{A}_x + r \tilde{D}_2 = D_{xx} \implies \tilde{A}_x = D_{xx} - r \tilde{D}_2, \quad (21)$$

$$\partial_{yy} : \quad \tilde{A}_y + \frac{1}{r} \tilde{D}_2 = D_{yy} \implies \tilde{A}_y = D_{yy} - \frac{\tilde{D}_2}{r}. \quad (22)$$

Substitute $D_{xy} = d_2, D_{xx} = (d_1 + d_2)/2, D_{yy} = d_2/2$:

$$\begin{aligned} \tilde{D}_2[i] &= \frac{D_{xy}[i]}{2} = \frac{d_2[i]}{2}, \\ \tilde{A}_x[i] &= D_{xx}[i] - r \tilde{D}_2[i] = \frac{d_1[i] + d_2[i]}{2} - \frac{rd_2[i]}{2} = \frac{d_1[i] + d_2[i](1-r)}{2}, \\ \tilde{A}_y[i] &= D_{yy}[i] - \frac{1}{r} \tilde{D}_2[i] = \frac{d_2[i]}{2} - \frac{d_2[i]}{2r} = \frac{d_2[i](r-1)}{2r}. \end{aligned}$$

Hence

$$\boxed{\tilde{D}_2[i] = \frac{d_2[i]}{2}, \quad \tilde{A}_x[i] = \frac{d_1[i] + d_2[i](1-r)}{2}, \quad \tilde{A}_y[i] = \frac{d_2[i](r-1)}{2r}.}$$

³Note that the $\mathcal{O}(h^3)$ terms have opposite signs in the two formulas and cancel. Only even orders remain.

$d_1[i]$ depends on C_{10} , which changes due to the source G_{10} and the absorption ϕ_{10} , so $\tilde{A}_x[i]$ must be recomputed every time step. $d_2[i]$ depends on C_{11} , which has no external source but can still change via $\phi_{11} \neq 0$ (net absorption/desorption of $(1, 1)$ particles). When the simulation runs with `fixed_mobile=true`, both concentrations are held constant and no recomputation is needed; otherwise both d_1 and d_2 are refreshed each step.

4.3 Comparison to old model

Quantity	Old model $\{(0, 1), (1, 1)\}$	New model $\{(1, 0), (1, 1)\}$
D_{xx}	$d/2$	$(d_1 + d_2)/2$
D_{yy}	d	$d_2/2$
D_{xy}	d	d_2
$\tilde{D}_2$	$\tilde{D}_2 = d/2$	$\tilde{D}_2 = d_2/2$
$\tilde{A}_x$	$\tilde{A}_x = D_{xx} - r\tilde{D}_2$	$\tilde{A}_x = \frac{d_1 + d_2(1-r)}{2}$
$\tilde{A}_y$	$\tilde{A}_y = D_{yy} - \tilde{D}_2/r$	$\tilde{A}_y = \frac{d_2(r-1)}{2r}$

5 Non-Negativity and Mesh Constraint

We need $\tilde{D}_2 \geq 0$, $\tilde{A}_x \geq 0$, $\tilde{A}_y \geq 0$, for us to invert the diffusion matrix later on as shown later in Chapter 7.

$$\begin{aligned}
\tilde{D}_2[i] \geq 0 &\iff d_2[i] \geq 0, \\
\tilde{A}_y[i] \geq 0 &\iff \frac{d_2[i](r-1)}{2r} \geq 0 \iff r \geq 1, \\
\tilde{A}_x[i] \geq 0 &\iff d_1[i] + d_2[i](1-r) \geq 0 \iff d_1[i] \geq d_2[i](r-1).
\end{aligned}$$

Define

$$\rho = \inf_{i,t} \frac{d_1[i,t]}{d_2[i,t]}.$$

Then a sufficient bound is

$$1 \leq r \leq 1 + \rho.$$

Reason for opposite direction vs old model: in this new mobile set the anisotropy is dominated by the x -axis contribution, so preserving non-negative $\tilde{A}_y$ requires $r \geq 1$ rather than $r \leq 1$.

6 Mass Conservation and Flux-Form Discretisation

All the parts in orange were just copied from last week's notes, for reference.

To enforce exact mass conservation the diffusion operator must be cast entirely in flux form:

$$\partial_t U + \nabla \cdot \mathbf{J} = 0, \quad \mathbf{J} = \mathbf{J}_{\text{adv}} + \mathbf{J}_{\text{diff}}. \quad (23)$$

Each face flux is computed once and applied with opposite signs in the two cells sharing that face, so internal contributions cancel in the global mass sum.

6.1 FV diffusion fluxes

From (18) we have:

$$\text{Diff}(C) = \partial_{xx}(\tilde{A}_x C) + \partial_{yy}(\tilde{A}_y C) + \tilde{D}_2 \mathcal{T}_h(C), \quad (24)$$

6.1.1 $\tilde{A}_x$ part

Write $\partial_{xx}(\tilde{A}_x C) = \partial_x G^x$ with

$$G^x = \partial_x(\tilde{A}_x C). \quad (25)$$

Face approximation at $(i + 1/2, j)$ with arithmetic coefficient averaging:

$$G_{i+1/2,j}^x = \tilde{A}_{x,i+1/2,j} \frac{U_{i+1,j} - U_{i,j}}{\Delta x}, \quad \tilde{A}_{x,i+1/2,j} = \frac{\tilde{A}_{x,i,j} + \tilde{A}_{x,i+1,j}}{2}. \quad (26)$$

Cell divergence:

$$\text{diff}_{i,j}^{(x)} = \frac{G_{i+1/2,j}^x - G_{i-1/2,j}^x}{\Delta x}. \quad (27)$$

Boundary fluxes are set to zero: $G_{1/2,j}^x = G_{N_x+1/2,j}^x = 0$.

6.1.2 $\tilde{A}_y$ part

Identical idea as in 6.1.1

Write $\partial_{yy}(\tilde{A}_y C) = \partial_y G^y$ with $G^y = \partial_y(\tilde{A}_y C)$. Face approximation at $(i, j + 1/2)$:

$$G_{i,j+1/2}^y = \tilde{A}_{y,i,j+1/2} \frac{U_{i,j+1} - U_{i,j}}{\Delta y}, \quad \tilde{A}_{y,i,j+1/2} = \frac{\tilde{A}_{y,i,j} + \tilde{A}_{y,i,j+1}}{2}. \quad (28)$$

Cell divergence:

$$\text{diff}_{i,j}^{(y)} = \frac{G_{i,j+1/2}^y - G_{i,j-1/2}^y}{\Delta y} \quad (29)$$

Boundary fluxes: $G_{i,1/2}^y = G_{i,N_y+1/2}^y = 0$.

6.1.3 Diagonal part.

The stencil (16) motivates the following corner flux (Treating diagonal like a usual finite difference approximation, but with Δx and Δy in denominator). At the north-east corner $(i + 1/2, j + 1/2)$, define

$$G_{i+1/2,j+1/2}^d = \tilde{D}_{2,i+1/2,j+1/2} (U_{i+1,j+1} - U_{i,j}), \quad \tilde{D}_{2,i+1/2,j+1/2} = \frac{\tilde{D}_{2,i,j} + \tilde{D}_{2,i+1,j+1}}{2}. \quad (30)$$

$$\text{diff}_{i,j}^{(d)} = \frac{G_{i+1/2,j+1/2}^d - G_{i-1/2,j-1/2}^d}{\Delta x \Delta y}. \quad (31)$$

Boundary corners with no physical neighbour are assigned $G^d = 0$ (zero-flux / reflecting condition along the diagonal direction).

7 Implicit Diffusion Matrix Assembly (IMEX)

Time splitting:

$$U^* = U^n + \Delta t \text{adv}(U^n), \quad (32)$$

$$(I - \Delta t L_{\text{diff}}) U^{n+1} = U^*. \quad (33)$$

Row-major index: $k = (i - 1)N_y + j$.

7.1 Interior row stencil

For interior (i, j) we define six face-averaged diffusion coefficients (reason on why they are defined like this follows shortly)

$$\tilde{a}_{i+1/2}^x = \frac{\tilde{A}_{x,i,j} + \tilde{A}_{x,i+1,j}}{2 \Delta x^2}, \quad \tilde{a}_{i-1/2}^x = \frac{\tilde{A}_{x,i,j} + \tilde{A}_{x,i-1,j}}{2 \Delta x^2}, \quad (34)$$

$$\tilde{a}_{j+1/2}^y = \frac{\tilde{A}_{y,i,j} + \tilde{A}_{y,i,j+1}}{2 \Delta y^2}, \quad \tilde{a}_{j-1/2}^y = \frac{\tilde{A}_{y,i,j} + \tilde{A}_{y,i,j-1}}{2 \Delta y^2}, \quad (35)$$

$$\tilde{d}_+ = \frac{\tilde{D}_{2,i,j} + \tilde{D}_{2,i+1,j+1}}{2 \Delta x \Delta y}, \quad \tilde{d}_- = \frac{\tilde{D}_{2,i,j} + \tilde{D}_{2,i-1,j-1}}{2 \Delta x \Delta y}. \quad (36)$$

Now we look at all the cells affected around a given cell. This is captured in L. **We discussed in the call that the matrix L "just has to be negative". But this cant be the case as the only the diagonal is constructed using negative numbers. Hence we discuss diagonal dominance, and not only negativity.**

$$L_{k,k} += -(\tilde{a}_{i+1/2}^x + \tilde{a}_{i-1/2}^x + \tilde{a}_{j+1/2}^y + \tilde{a}_{j-1/2}^y + \tilde{d}_+ + \tilde{d}_-), \quad (37)$$

$$L_{k,k(i+1,j)} += \tilde{a}_{i+1/2}^x, \quad L_{k,k(i-1,j)} += \tilde{a}_{i-1/2}^x, \quad (38)$$

$$L_{k,k(i,j+1)} += \tilde{a}_{j+1/2}^y, \quad L_{k,k(i,j-1)} += \tilde{a}_{j-1/2}^y, \quad (39)$$

$$L_{k,k(i+1,j+1)} += \tilde{d}_+, \quad L_{k,k(i-1,j-1)} += \tilde{d}_-. \quad (40)$$

When inserting the upper equations into the L equations we can recover exactly diff-notations (27), (29), and (31) (when looking at what cells are impacted in diff formulation).

7.2 Diagonal dominance of $A = I - \Delta t L$

Non-negativity and mesh-aspect-ratio constraint. Provided the mesh satisfies (5), all coefficients $\tilde{A}_x, \tilde{A}_y, \tilde{D}_2 \geq 0$, which ensures:

$$L_{kk} \leq 0, \quad L_{k\ell} \geq 0 \ (\ell \neq k), \quad \sum_{\ell} L_{k\ell} = 0. \quad (41)$$

For $A = I - \Delta t L$:

$$A_{kk} = 1 + \Delta t |L_{kk}|, \quad A_{k\ell} = -\Delta t L_{k\ell} \leq 0 \ (\ell \neq k), \quad (42)$$

and

$$A_{kk} - \sum_{\ell \neq k} |A_{k\ell}| = 1 + \Delta t |L_{kk}| - \Delta t \sum_{\ell \neq k} L_{k\ell} = 1 + \Delta t (|L_{kk}| - \sum_{\ell \neq k} L_{k\ell}) = 1 + \Delta t (\sum_{\ell} L_{k\ell}) = 1 > 0. \quad (43)$$

Hence A is strictly diagonally dominant by (hence non-singular), guaranteeing a unique solution to the linear system.

7.3 Advection

Advection is treated explicitly in conservative flux form using MP5 reconstruction (5th-order monotonicity-preserving) with SSP-RK3 time integration. The drift coefficients $F_x[i]$ and $F_y[i]$ depend only on x , not on y (see Section 3.2), so each row/column sweep uses the same velocity vector for all cells in that row/column.

In the IMEX variant, a Strang-splitting is used: half-step advection (MP5+RK3), full implicit diffusion solve, then another half-step advection. This keeps diffusion unconditionally stable while the advection CFL governs the step size.

8 CFL and Stability

Exact steps shown in last week's document, here just a summary: Explicit update uses

$$\Delta t_{adv} = \frac{1}{\|F_x\|_\infty/\Delta x + \|F_y\|_\infty/\Delta y},$$

$$\Delta t_{diff} = \frac{1/2}{\|\tilde{A}_x\|_\infty/\Delta x^2 + \|\tilde{A}_y\|_\infty/\Delta y^2 + \|\tilde{D}_2\|_\infty/(\Delta x \Delta y)},$$

$$\Delta t = \sigma \min(\Delta t_{adv}, \Delta t_{diff}).$$

IMEX uses advection CFL only:

$$\Delta t = \sigma \Delta t_{adv},$$

with implicit diffusion solve $(I - \Delta t L^{n+1})U^{n+1} = U^*$. Since C_{10}, C_{11} evolve when $G_{10} > 0$, coefficients and CFL bounds are recomputed every step.

9 Did Strang-splitting solve our mass-drift problem?

9.1 Simplified old problem formulation $\{(0, 1), (1, 1)\}$

To avoid small mass-drift, we earlier did a mass-drift correction (as outlined in last week's document). In the call, we discussed that the small mass-drift might be resolved using a more precise time-stepping scheme. The theory was that the error was introduced by our upwind-scheme. We hence changed to strang-splitting in our IMEX method (half step advection, full step diffusion, half step advection) and started using a higher-order advection method. Advection is now calculated using MP5 reconstruction and a RK3 scheme.

Using the mass-correction looks like depicted in Fig. 1. Fig. 2 shows the IMEX method without correction before increasing the order of the method.

After changing to using strang-splitting, MP5, and RK3, we get 3. **The results do not look better than before, which is surprising. We did not yet find the reason for this problem.**

Figure 1: Relative total mass drift on entire domain over time for six different tests with Finite Volume IMEX method (purely explicit method looks almost identical). Uses mass-correction.

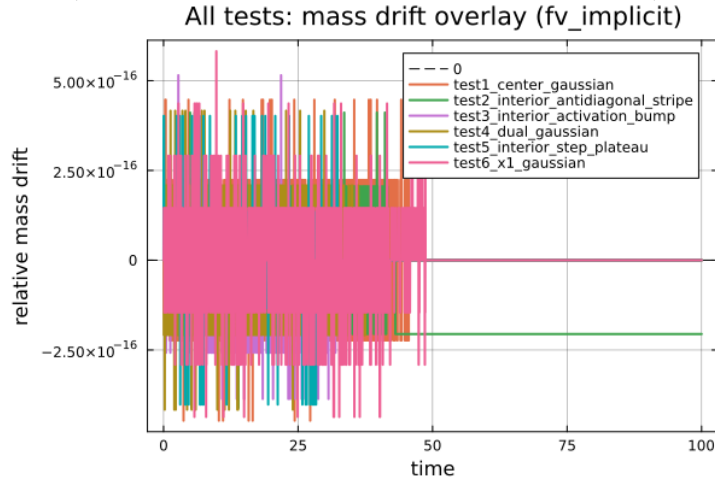


Figure 2: Relative total mass drift on entire domain over time for six different tests with Finite Volume IMEX method. Does **not** use mass-correction. [Old method; lower order](#)

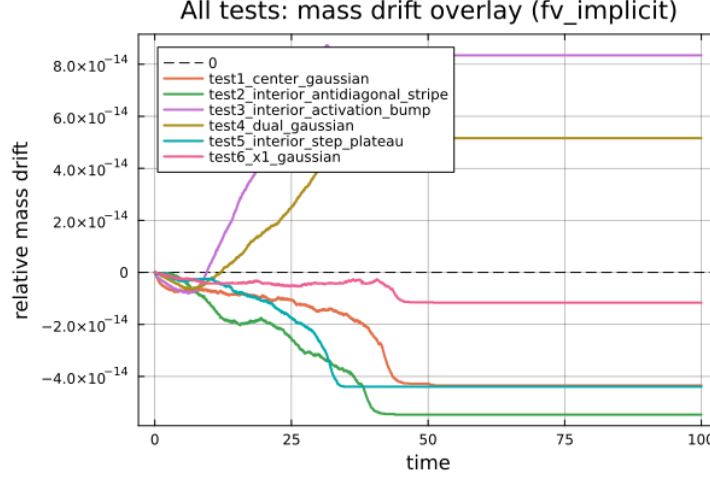
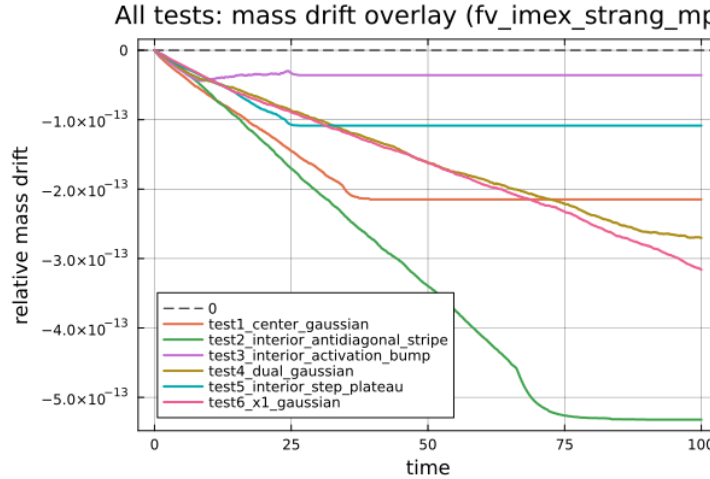


Figure 3: Relative total mass drift on entire domain over time for six different tests with Finite Volume IMEX method. Does **not** use mass-correction. [Strang-splitting, MP5, RK3](#)

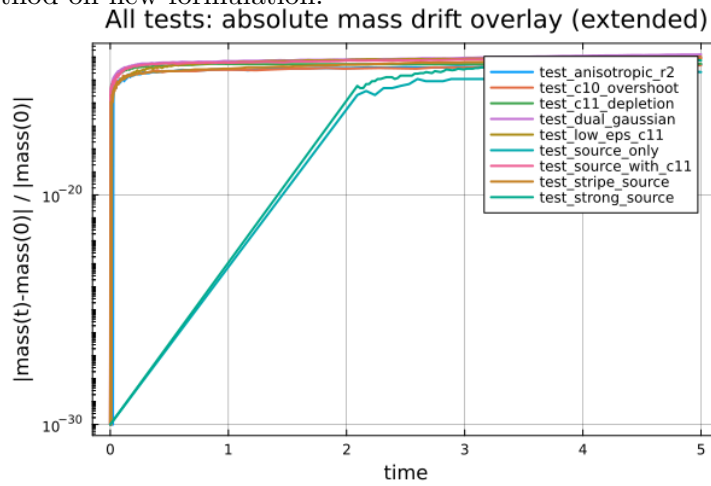


9.2 New model $\{(1, 0), (1, 1)\}$

The new model almost had no mass drift. We assume that this is due to handling the mass-evolutions of $(0, 1)$ and $(1, 0)$ explicitly in the method. Before we just ignored that these values change over time.

In Fig. 4 we can observe that mass does not drift anymore. Note that these now use different experiment-setups than before, because here we tested for the impact of source-terms and initial concentrations. Nevertheless, we can conclude that this method is most likely generally stable regarding the mass.

Figure 4: Relative total mass drift on entire domain over time for six different tests with Finite Volume IMEX method on new formulation.



10 Numerical Experiments

Please note that the attached plots were generated with the fully explicit method. The IMEX scheme produced identical results.

10.1 Test 1 — Reference: moderate source, no C_{11}

Establishes the baseline: a centre Gaussian drifts purely in the x -direction driven by G_{10} . No y -dynamics occur because $C_{11} \equiv 0$.

Parameter	Value
IC	Centre Gaussian, domain $x_{\max} = 90$, $y_{\max} = 36$
ε	1.0
G_{10}	2.0
C_{10}^0	0.01
C_{11}^0	0.0

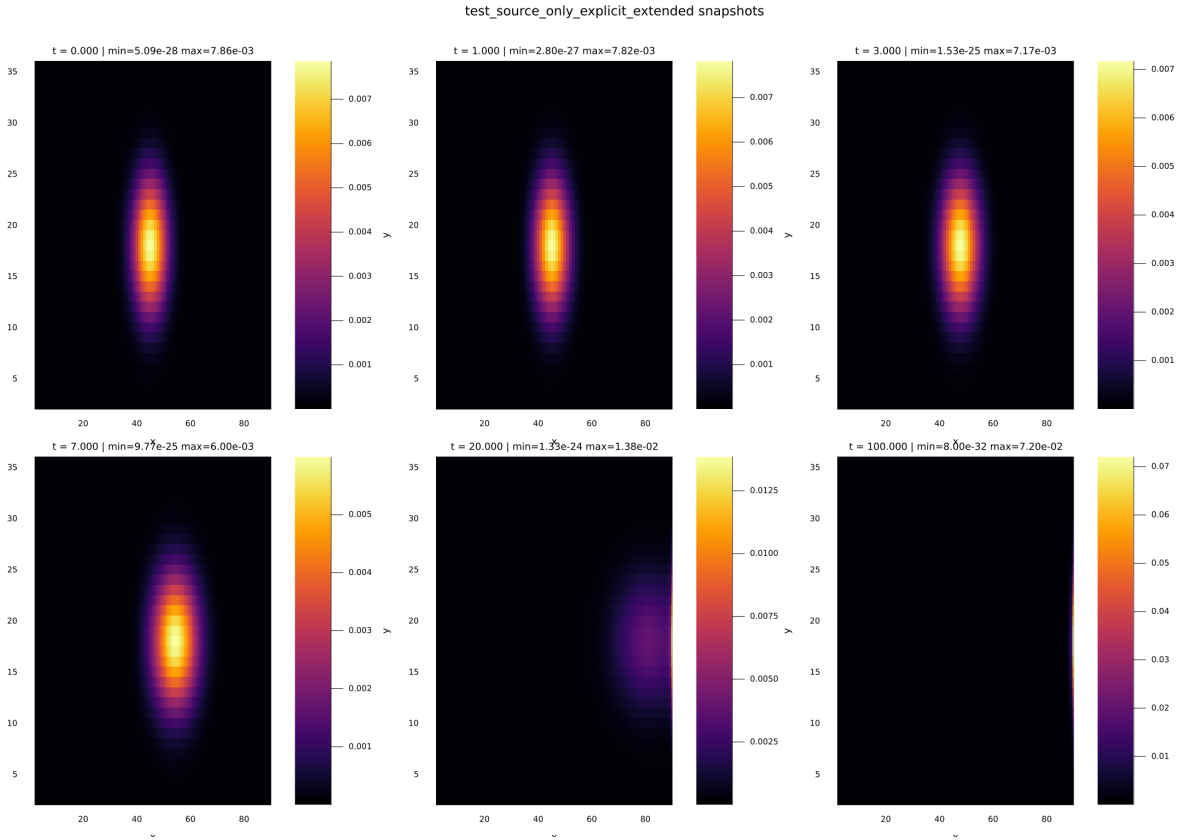


Figure 5: Test 1 (explicit): centre Gaussian under moderate source $G_{10} = 2$, $T = 100$.

10.2 Test 2 — Strong source: fast x-drift

Triples the source rate.

Parameter	Value
IC	Centre Gaussian, domain $x_{\max} = 90$, $y_{\max} = 36$
ε	1.0
G_{10}	6.0
C_{10}^0	0.01
C_{11}^0	0.0

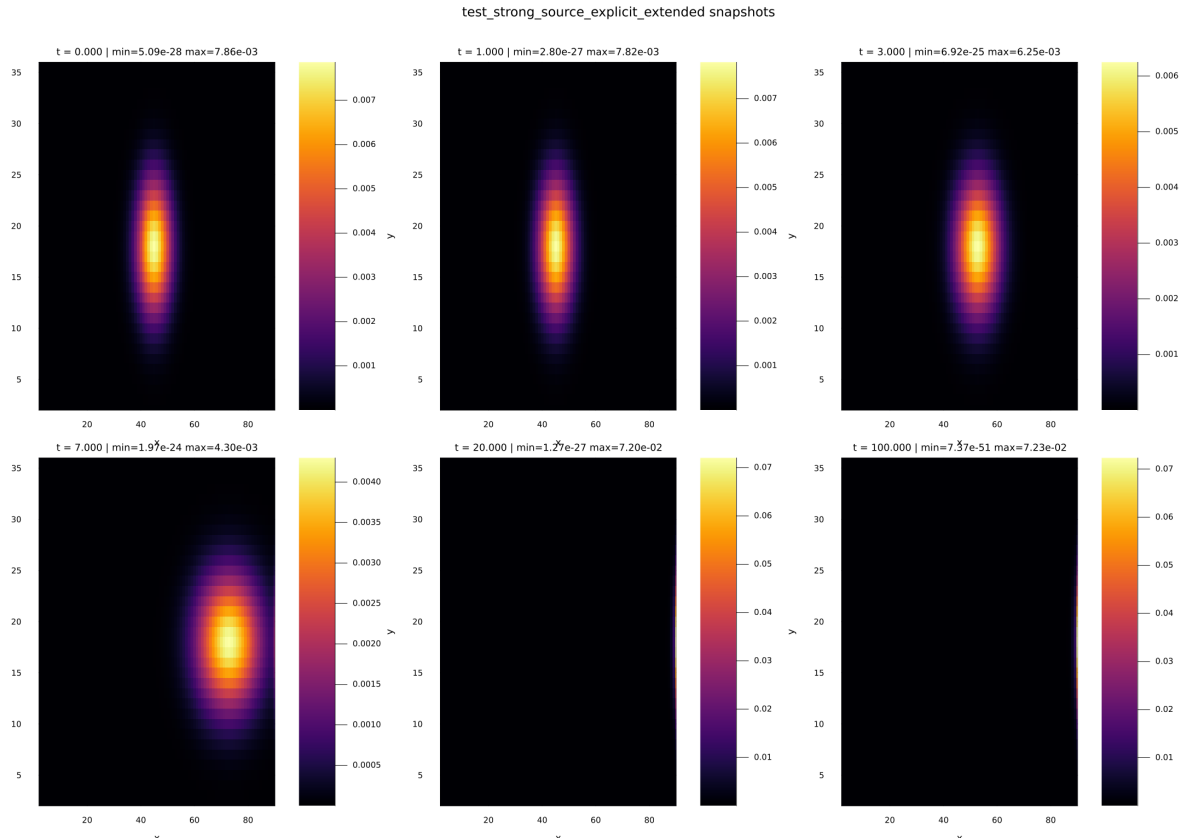


Figure 6: Test 2 (explicit): strong source $G_{10} = 6$, $T = 100$.

10.3 Test 3 — C_{10} overshoot: transient then quasi-steady drift

Starts with $C_{10}^0 = 5$. The fast concentration gives a fast initial drift.

Parameter	Value
IC	Centre Gaussian, domain $x_{\max} = 90$, $y_{\max} = 36$
ε	1.0
G_{10}	1.0
C_{10}^0	5.0
C_{11}^0	0.0

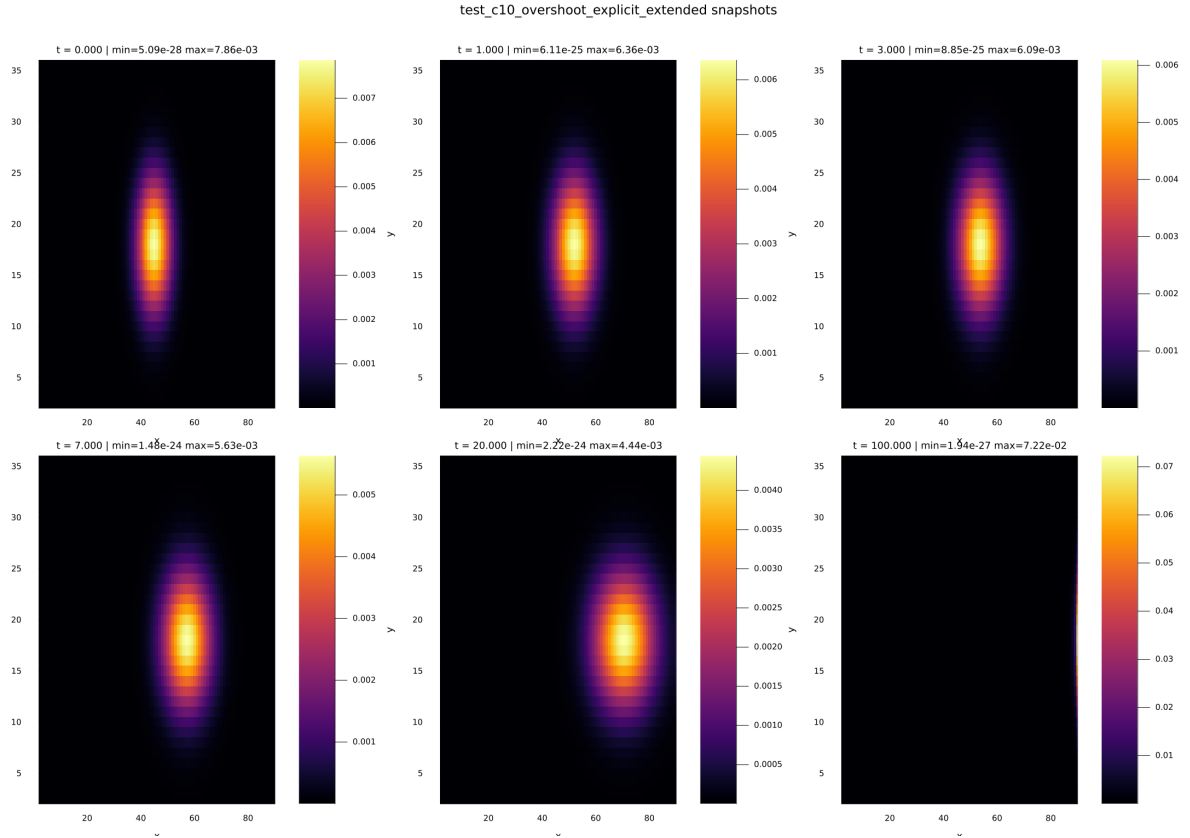


Figure 7: Test 3 (explicit): $C_{10}^0 = 5$ overshoot with $G_{10} = 1$, $T = 100$.

10.4 Test 4 — Source with initial C_{11} : combined x+y dynamics

Adds $C_{11}^0 = 5$ to the moderate-source run. Because C_{11} depletes on a timescale $\tau_{11} \approx 1/B_{11} \ll 1$, the cluster experiences a brief ≈ 5 -unit y-shift immediately followed by sustained G_{10} -driven x-drift.

Parameter	Value
IC	Centre Gaussian, domain $x_{\max} = 90$, $y_{\max} = 36$
ε	1.0
G_{10}	2.0
C_{10}^0	0.01
C_{11}^0	5.0

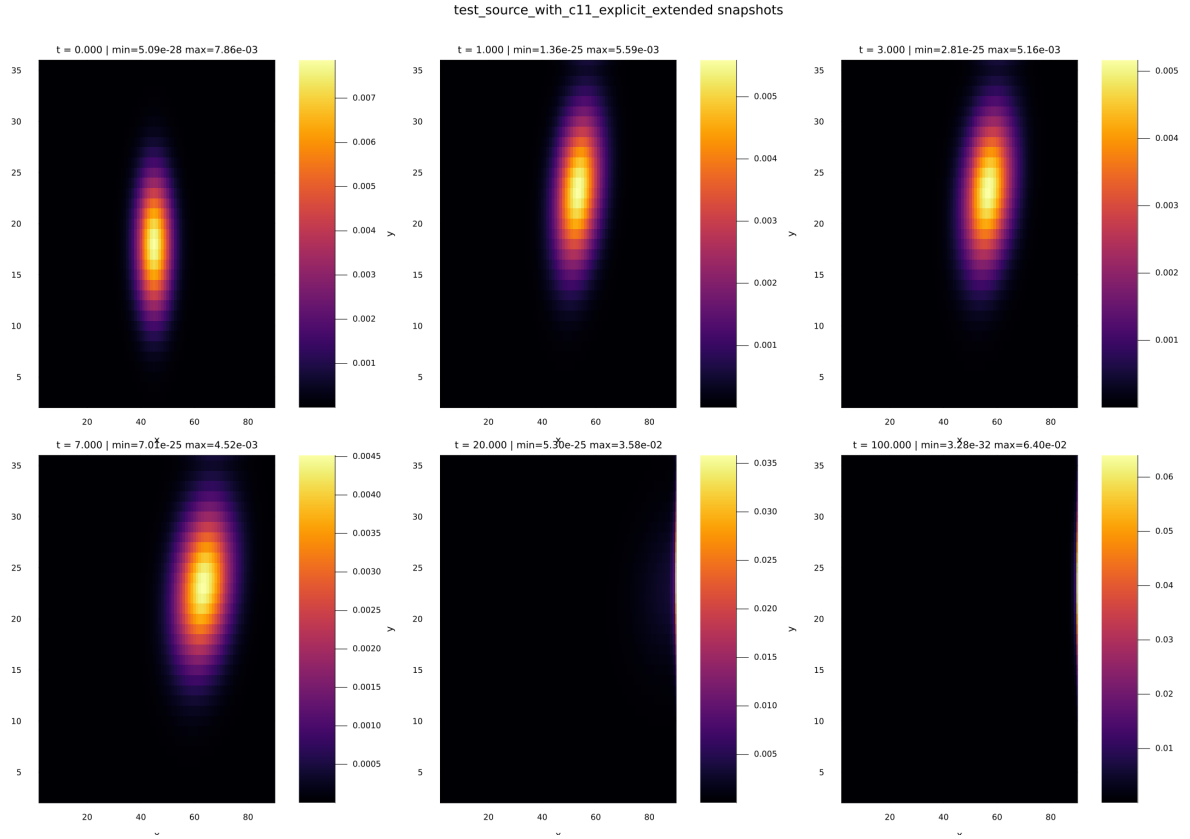


Figure 8: Test 4 (explicit): moderate source $G_{10} = 2$ plus $C_{11}^0 = 5$, $T = 100$.

10.5 Test 5 — Low ε : slower C_{11} depletion

Same as Test 4 except $\varepsilon = 0.3$. Since $B_{11} \propto \varepsilon$, the depletion timescale $\tau_{11} = 1/B_{11}$ is $\approx 3\times$ longer. The total y-shift ($\approx C_{11}^0 = 5$ units) is identical, but it is spread over a longer time interval.

Parameter	Value
IC	Centre Gaussian, domain $x_{\max} = 90$, $y_{\max} = 36$
ε	0.3
G_{10}	2.0
C_{10}^0	0.01
C_{11}^0	5.0

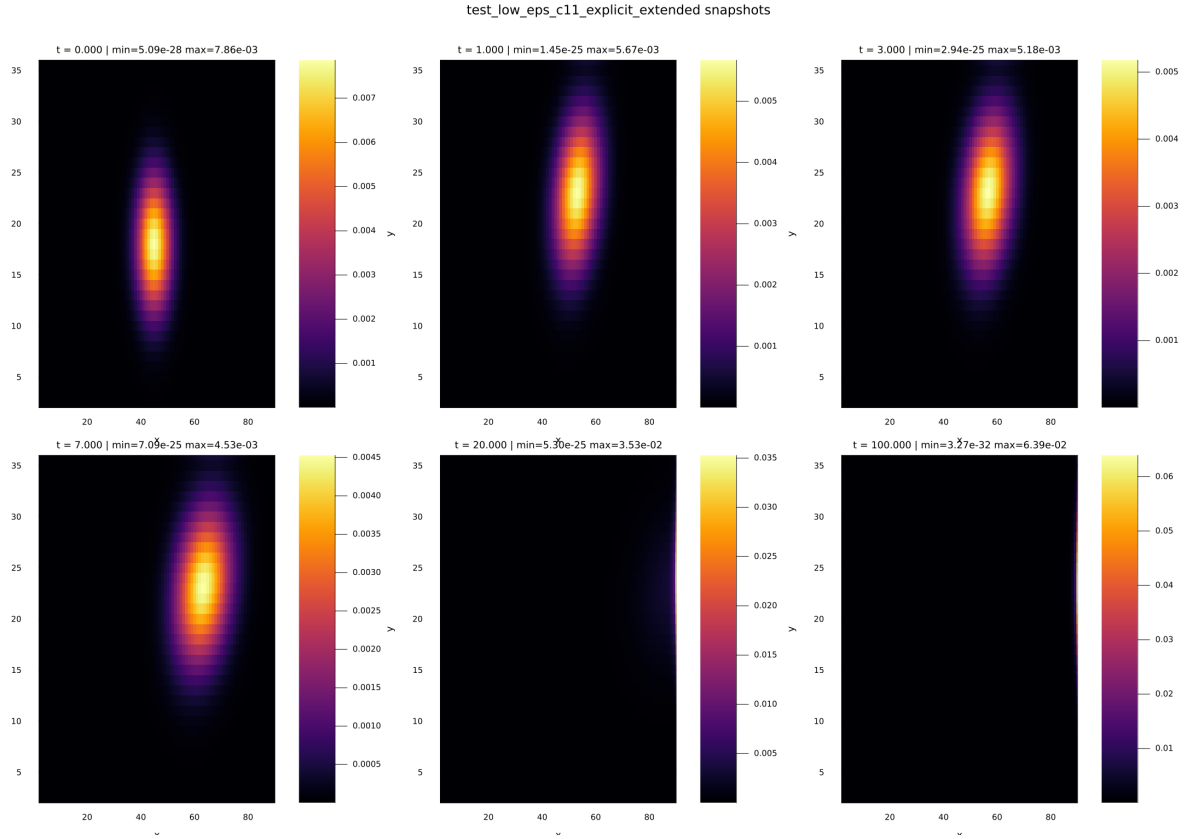


Figure 9: Test 5 (explicit): $\varepsilon = 0.3$ with $C_{11}^0 = 5$, $G_{10} = 2$, $T = 100$.

10.6 Test 6 — C_{11} depletion only ($G_{10} = 0$)

The only test with $G_{10} = 0$. No x -source means no sustained x -drift; the cluster shifts upward as C_{11} depletes, then stops.

Parameter	Value
IC	Centre Gaussian, domain $x_{\max} = 90$, $y_{\max} = 36$
ε	1.0
G_{10}	0.0
C_{10}^0	0.0
C_{11}^0	5.0

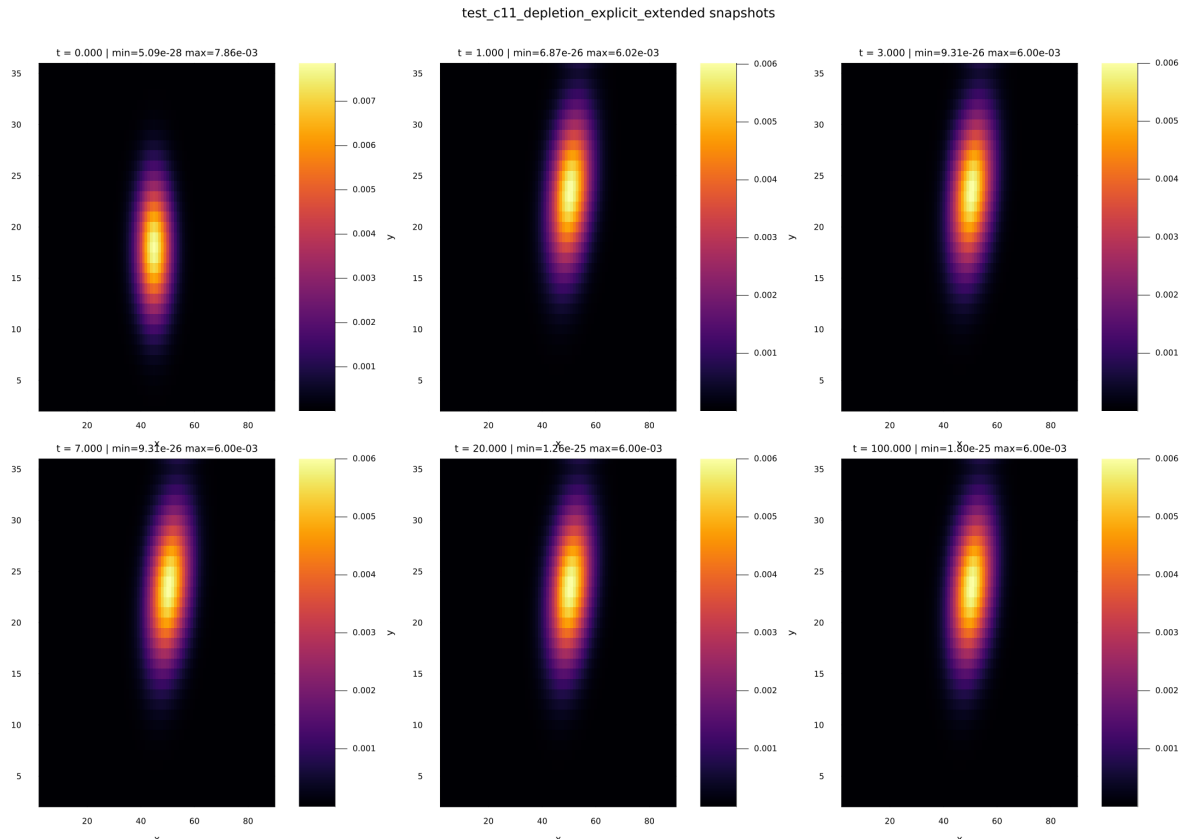


Figure 10: Test 6 (explicit): pure C_{11} depletion, $G_{10} = 0$, $C_{11}^0 = 5$, $T = 100$.

10.7 Test 7 — Stripe IC under strong source with C_{11}

A different initial condition: an anti-diagonal stripe envelope. The stripe is initially tilted; the combined x-drift ($G_{10} = 3$) and y-shift ($C_{11}^0 = 2$) deform the stripe in a qualitatively different way from a Gaussian.

Parameter	Value
IC	Anti-diagonal stripe, domain $x_{\max} = 108$, $y_{\max} = 42$
ε	1.0
G_{10}	3.0
C_{10}^0	0.01
C_{11}^0	2.0

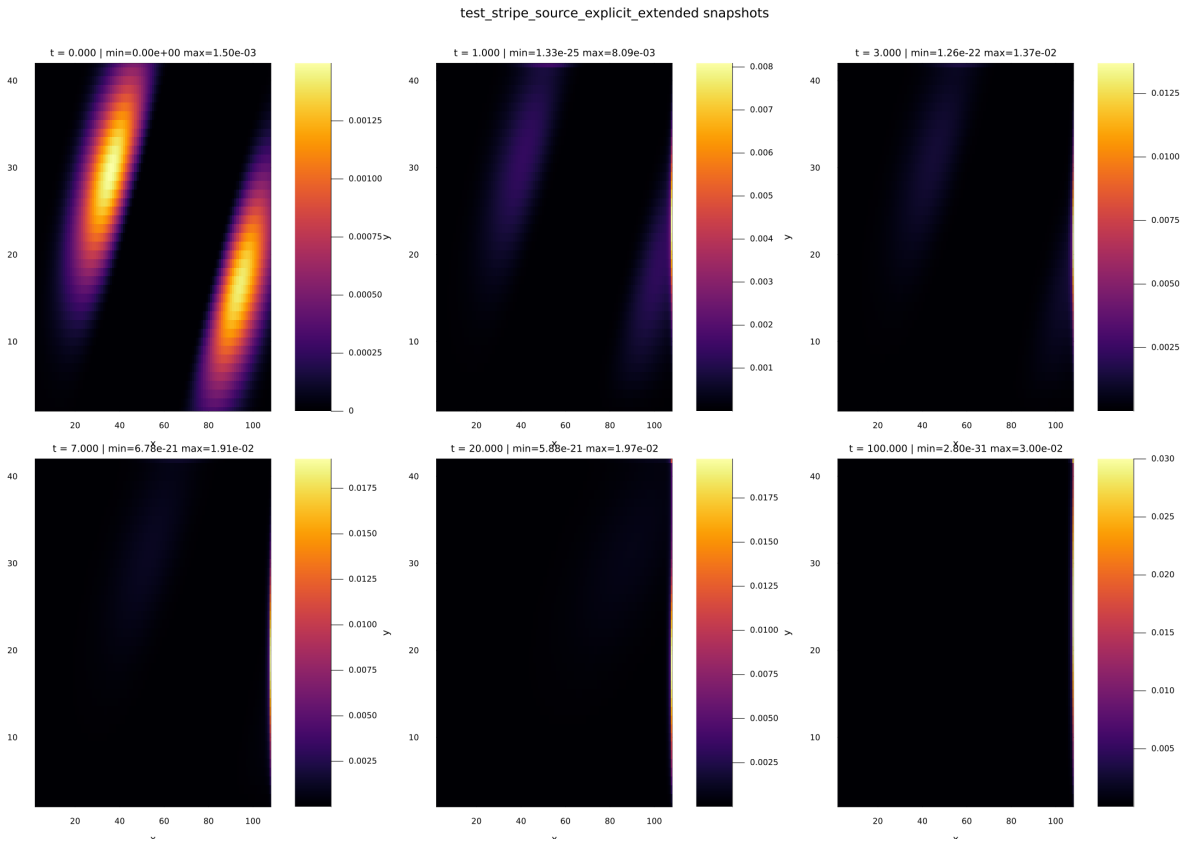


Figure 11: Test 7 (explicit): anti-diagonal stripe with $G_{10} = 3$, $C_{11}^0 = 2$, $T = 100$.

10.8 Test 8 — Dual Gaussian with high ε and combined drift

Two Gaussian blobs at $(x, y) \approx (15, 0.35 y_{\max})$ and $(34, 0.75 y_{\max})$. The right blob at larger x experiences a faster x-drift (since $F_x \propto C_{10}\sqrt{x}$), so the two blobs progressively separate in x .

Parameter	Value
IC	Dual Gaussian, domain $x_{\max} = 144, y_{\max} = 54$
ε	1.5
G_{10}	1.5
C_{10}^0	0.01
C_{11}^0	3.0

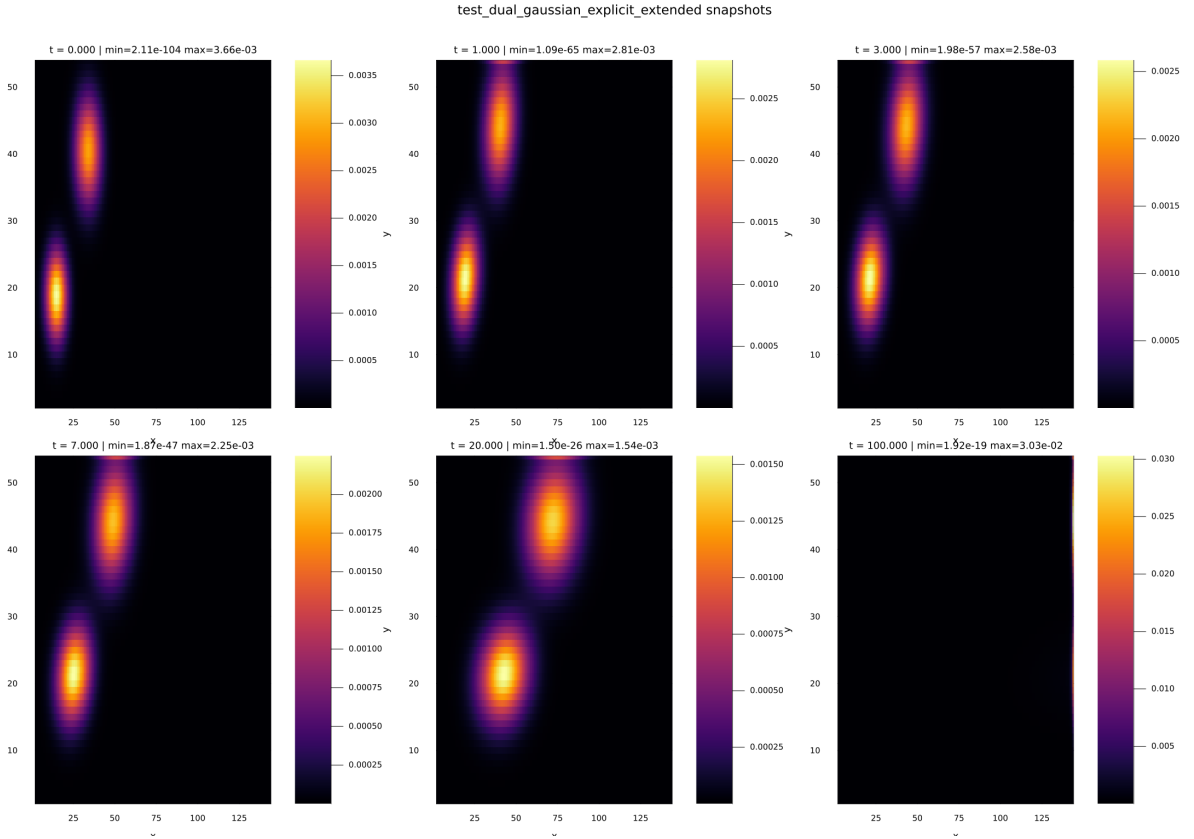


Figure 12: Test 8 (explicit): dual Gaussian with $\varepsilon = 1.5$, $G_{10} = 1.5$, $C_{11}^0 = 3$, $T = 100$.

10.9 Test 9 — Anisotropic grid ($r = 2$, $\Delta x = 2 \Delta y$)

All previous tests used $r = \Delta x / \Delta y = 1$. This test validates the mesh-adapted splitting for $r > 1$. With $r = 2$ the y -grid is twice as fine as the x -grid ($\Delta x = 1$, $\Delta y = 0.5$, $N_y = 69$ instead of 35 on the same physical domain).

The non-negativity constraint requires $r \leq 1 + \rho$ where $\rho = \inf_i d_1[i] / d_2[i]$. At $t = 0$ with $C_{10}^0 = 2$, $C_{11}^0 = 1$, $\varepsilon = 1$: $\rho \approx C_{10} / C_{11} = 2$, so $r = 2 \leq 1 + 2 = 3$ is satisfied with margin. As C_{11} depletes the ratio increases further, so the constraint is never violated.

Parameter	Value
IC	Centre Gaussian, domain $x_{\max} = 90$, $y_{\max} = 36$
Grid	$N_x = 89$, $N_y = 69$; $\Delta x = 1$, $\Delta y = 0.5$; $r = 2$
ε	1.0
G_{10}	1.0
C_{10}^0	2.0
C_{11}^0	1.0

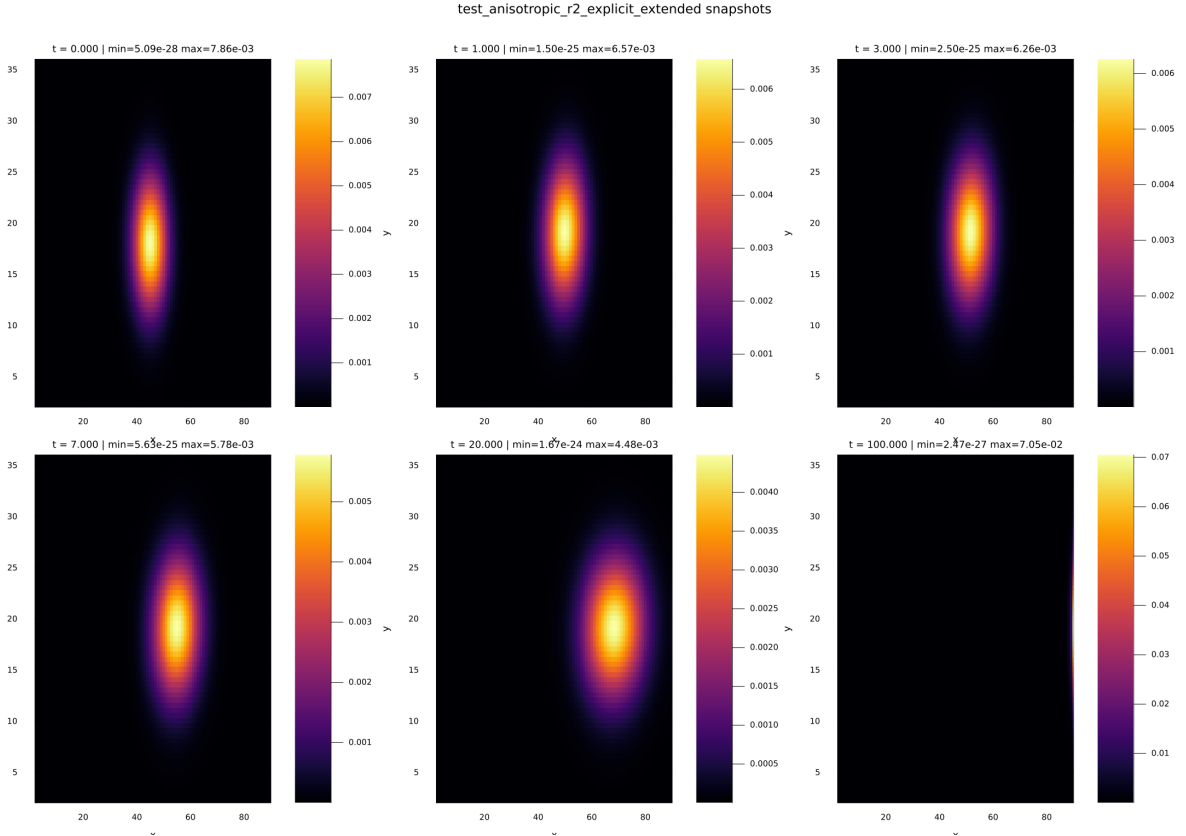


Figure 13: Test 9 (explicit): anisotropic grid $r = 2$, $C_{10}^0 = 2$, $C_{11}^0 = 1$, $G_{10} = 1$, $T = 100$.

The figures below show line-quantity overlays for all eight tests run with the explicit solver. The IMEX overlay figures are analogous.

11 Diagnostic overlays

The figures below show line-quantity overlays for all eight tests run with the explicit solver. The IMEX overlay figures are analogous.

extended_model: line overlays

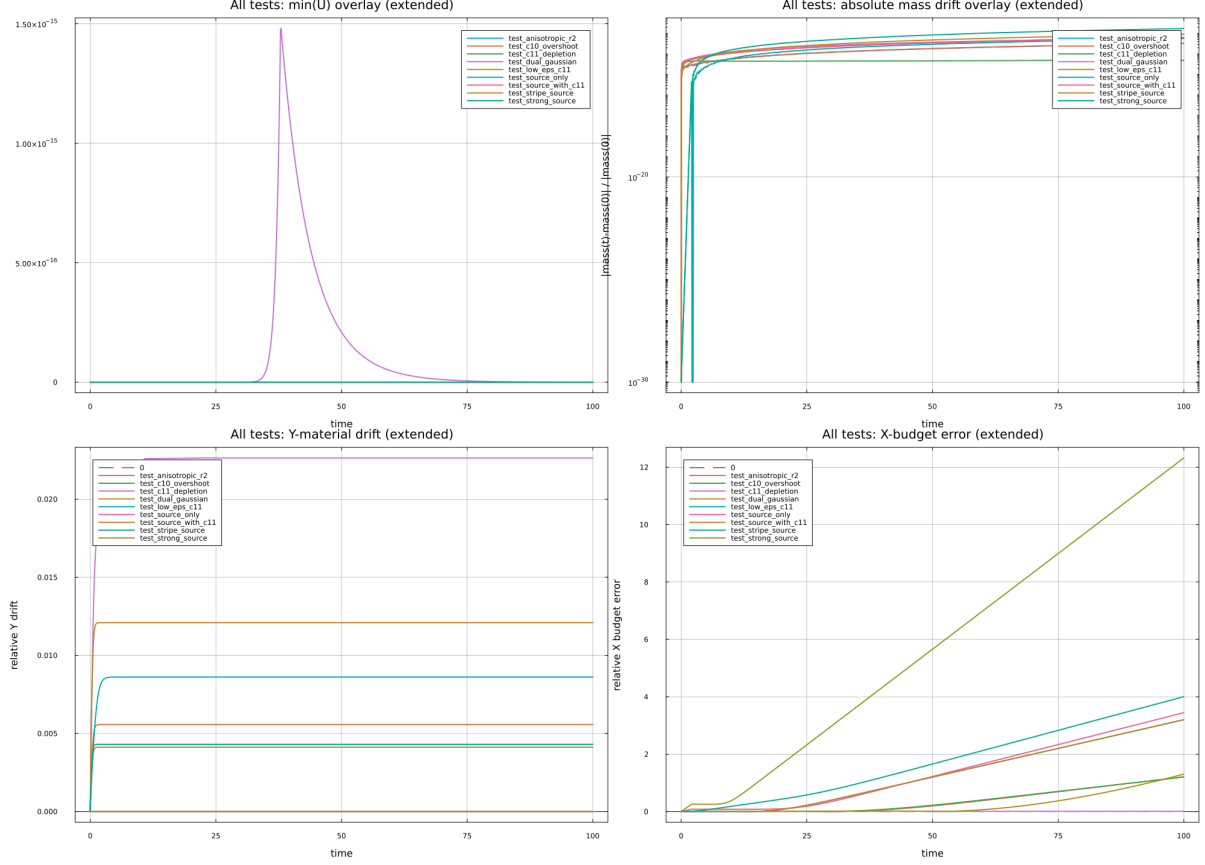


Figure 14: All tests (explicit): $\min(U)$, relative mass drift δ_M , relative Y_{mat} drift (expect this to stay constant), and relative X_{mat} budget error (here we expect the source term contribution, which we subtracted here before).

extended_model: mobile concentrations

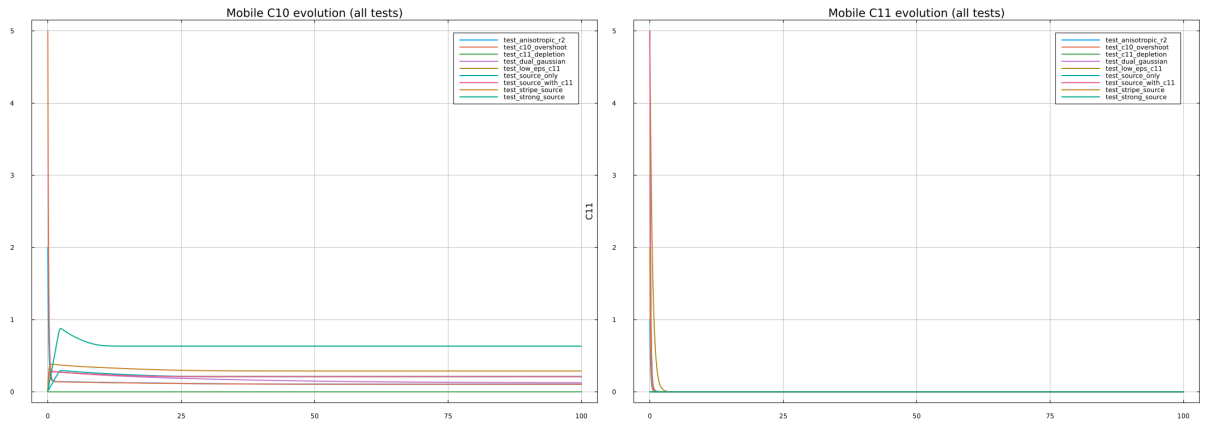


Figure 15: All tests (explicit): $C_{10}(t)$ and $C_{11}(t)$ for each experiment. Tests with $C_{11}^0 > 0$ show a rapid depletion spike; tests with $G_{10} > 0$ show C_{10} rising to quasi-steady state.

Discussion of the four diagnostic panels

$\min(U)$ (top left). We see that we keep our concentrations positive. Only one line briefly rises, but this is within very small ranges (order of machine precision). The most important thing is that no negative concentrations are ever produced.

Mass drift (top right). All nine tests track almost no mass drift. This is a lot better than with the old model.

Y_{mat} drift (bottom left). Section 2.3 shows that $Y_{\text{mat}} = C_{11} + \iint y U dx dy$ is exactly conserved by the *continuous* equations. This is also the case in our numerical experiments, after some time for the monomers to find their final concentration.

X_{mat} budget error (bottom right). Section 2.3 shows $\dot{X}_{\text{mat}} = G_{10}$ exactly. Here we calculated the X_{mat} that we would expect (after adding the source contribution) and subtracted what we actually got. We can see that these values also stabilize in the beginning, but at some point start constantly rising. My theory is, that this is the case when after a certain time all the mass is trapped in the top-right corner of our system (no-flux boundary). Then the mass-addition maybe does not work properly on the corner values. I don't think is is a big issue for now.